



Disentangling the roles of age and insect colonization in the temporal succession of VOC profiles in rat decomposition

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Abstract

Accurately estimating the postmortem interval (PMI) is a pivotal forensic challenge, as the precision of “chemical clocks” based on volatile organic compounds (VOCs) is confounded by intrinsic factors like age and extrinsic factors like insect colonization. This study aimed to disentangle these effects using TD-GC-MS with multivariate analysis to profile VOCs from rats of three ages with and without *Sarcophaga peregrina* (Robineau-Desvoidy, 1830) (Diptera: Sarcophagidae) colonization. While decomposition time was the principal driver of VOC succession, exhibiting stage-specific signatures like early-phase 2-hexanone and mid-phase aromatics, age and insects modulated this timeline via distinct mechanisms. Age increased the chemical complexity of VOC profiles, with adult rats showing significantly more diverse VOC profiles than younger cohorts during days 10–14 ($p < 0.05$). In contrast, insect colonization acted as a potent accelerator, advancing the chemical timeline by approximately 6 days, with the day 4 profile of colonized rats mirroring the day 10 profile of insect-free ones. We demonstrate that age increases the chemical heterogeneity of VOC profiles while insects accelerate temporal progression. Both factors must therefore be integrated as critical variables in VOC-based PMI models to ensure accuracy.

Keywords Postmortem interval · Volatile organic compounds · Age effect · Insect colonization · TD-GC-MS

Introduction

Accurately estimating the postmortem interval (PMI) remains one of the central challenges in forensic practice [1]. In the later stages of decomposition, forensic entomology provides unique perspectives and critical evidence for the minimum postmortem interval (PMI_{min}) estimation

[2]. Estimates of PMI_{min} can be determined by aging the oldest immature necrophagous insect associated with the remains and then situating that estimate within the species’ expected colonization timeline under the documented environmental conditions. Necrophagous insects locate and colonize remains by detecting characteristic volatile organic compounds (VOCs) released during decomposition, and their successional patterns follow predictable successional sequences that have been shown to be useful for PMI_{min} estimation [3–6]. The chemical basis of this process lies in the dynamic evolution of VOC profiles over time as a body decomposes [7–9]. Thus, elucidating the temporal dynamics of VOCs and identifying the key intrinsic and extrinsic factors that shape these patterns not only deepens our understanding of the biochemical mechanisms underlying decomposition but also establishes a theoretical foundation for developing more accurate PMI estimation models and enhancing the evidential value of forensic entomology.

Corpse decomposition fundamentally involves the degradation of biological macromolecules—including lipids, proteins, and carbohydrates—through the combined action of microorganisms and endogenous enzymes,

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generating a wide spectrum of VOCs such as alkanes, aromatics, alcohols, sulfides, ketones, aldehydes, esters, and nitrogen-containing compounds [10, 11]. A milestone in this field was established by Vass et al. (2004), who systematically analyzed VOCs from human remains and compiled a database of 424 compounds spanning eight major classes based on functional groups, thereby laying the foundation for subsequent studies [12]. Since then, most research has focused on how external environmental variables shape decomposition VOC profiles, including temperature [8], humidity [8], seasonal effects [13], soil conditions [14], habitat differences [4], sorbent materials [15], and analytical methodologies [16]. By contrast, intrinsic biological factors—such as age, sex, and body mass [17]—have received comparatively little attention [18]. Recent studies indicate that human skin volatiles exhibit age-dependent changes that can be modeled through multiple regression analysis [19], and VOCs from excreta can discriminate between age groups [20]. However, within the domain of thanatochemistry, the relationship between carcass age and decomposition VOC profiles remains insufficiently explored, and the potential forensic utility of these age-related VOCs has yet to be fully realized.

Beyond intrinsic factors, insect colonization represents a critical driver of corpse decomposition. Necrophagous flies profoundly accelerate the breakdown of remains through larval feeding and associated biological activities, thereby reshaping both the pace and the temporal succession of decomposition VOC profiles [21, 22]. Sarcophagidae comprises approximately 3000 described species worldwide, and its ecological habits are closely associated with human life. *Sarcophaga peregrina* (Robineau-Desvoidy, 1830) (Diptera: Sarcophagidae) is one of the most widely distributed flesh flies, occurring from tropical to subtropical zones of the Palearctic, Oriental, and Oceanian regions [23]. Building on these insights, the present study aims to clarify how age (as a key intrinsic variable) and insect colonization (as a major extrinsic factor) jointly influence the temporal dynamics of VOC release during corpse decomposition. We employed gas chromatography-mass spectrometry coupled with thermal desorption (TD-GC-MS) to systematically profile VOC succession over a 20-day decomposition period in rats of three developmental stages (3, 5, and 8 weeks), both with and without colonization by *S. peregrina*. Multivariate approaches were integrated to disentangle the specific contributions of age and insect activity to VOC evolution. Through this framework, we seek to reveal the underlying mechanistic pathways by which intrinsic and extrinsic factors shape decomposition chemistry, thereby providing new empirical evidence to advance the precision of PMI estimation.

Materials and methods

All animal experiments were reviewed and approved by the Institutional Review Board of Xiangya School of Medicine, Central South University (Approval No. 2023-KT085). All procedures for animal housing, handling, and euthanasia complied strictly with institutional and national regulations on laboratory animal welfare.

Experimental animals and decomposition conditions

A total of 12 healthy male Sprague-Dawley (SD) rats were used in this study. Animals were assigned to three age groups: 3 weeks old ($n=3$, designated 3WM), 5 weeks old ($n=3$, designated 5WM), both of which served as insect-free groups, and 8 weeks old ($n=6$). The 8-week-old group was further randomly divided into two subgroups: an insect-free control group ($n=3$, designated 8WM) and an insect-colonized group ($n=3$, designated 8WI). The mean body mass was 63.3 ± 4.2 g for 3WM, 73.4 ± 3.7 g for 5WM, and 151.9 ± 4.1 g for 8-week-old rats (8WM and 8WI combined). The insect colonization was applied only to the 8-week-old group to isolate the effects of insect activity. For this, *S. peregrina* larvae were introduced to the 8WI group at the beginning of the decomposition process, as described in the “[Insect ect. rearing and colonization](#)” section.

All rats were euthanized by blunt-force cranial trauma and placed individually in polypropylene containers (17 cm $\times$ 10 cm $\times$ 6 cm) lined with sterile sand. Each container was covered with fine mesh to prevent external insect access while allowing passive air exchange with the chamber environment, thereby preventing the accumulation of volatiles and maintaining aerobic decomposition conditions between sampling intervals. All containers were placed in a single climate-controlled chamber (model GPL-350D, 60 cm $\times$ 56 cm $\times$ 105 cm; Labotery, China) equipped with a continuous internal air-circulation system operating at approximately 1 L/min. Before use, the interior was wiped with 75% ethanol and ventilated for 24 h. The chamber was maintained at 25 ± 1 °C, $70 \pm 5\%$ relative humidity, and a 12-h:12-h light-dark cycle to ensure standardized decomposition conditions. The temperature and humidity were chosen to approximate the local subtropical climate during the active season of *S. peregrina* and to support larval survival. Decomposition was carried out at 25 ± 1 °C, and sampling was performed at 0, 4, 10, 14, and 20 days postmortem. For comparability with previous studies, these time points correspond to approximately 0, 100, 250, 350, and 500 accumulated degree days (ADD, base 0 °C), respectively.

Insect rearing and colonization

The necrophagous fly *S. peregrina* was collected in the field in Changsha, Hunan Province, China, and reared under laboratory conditions (25 °C, 70% relative humidity, 12-h:12-h light-dark cycle) for more than five generations to establish a standardized colony. After mating, freshly deposited first-instar larvae (< 1 h) were collected from oviposition substrates (fresh pig lung). A total of 270 healthy larvae were randomly selected and evenly inoculated onto three 8-week-old rat carcasses (designated as the 8-week insect-colonized group, 8WI), with approximately 90 larvae introduced to the oral cavity, auditory canals, and anus of each carcass. Containers were subsequently covered with fine mesh to prevent larval escape.

VOC sampling and pre-treatment

VOCs were collected by dynamic airflow sampling of the headspace above the same individual animal in each group on decomposition days 0, 4, 10, 14, and 20, adapted from the protocol of Stefanuto et al. [16]. For each sampling event, the climate-controlled chamber was briefly opened, and the polypropylene container was placed inside a polytetrafluoroethylene (PTFE) sampling bag. The bag was then sealed and left inside the climate-controlled chamber for 30 min to allow VOCs to accumulate in the headspace under the controlled temperature and humidity (25 ± 1 °C, 70 ± 5% relative humidity). The sampling setup utilized a dynamic airflow headspace sampling configuration, as illustrated in Figure S1. Ambient air was first drawn through an activated carbon filter tube connected to the inlet of the PTFE bag to remove background contaminants. The filtered air then flowed over the carcass, and the VOC-enriched air in the headspace exited via the outlet of the PTFE bag, which was connected directly to the Tenax TA stainless-steel sorbent tube and subsequently to the QC-1S constant-flow air pump (Beijing Institute of Labour Protection, China). Thus, decomposition VOCs did not pass through the activated carbon at any point in the sampling line.

Sampling was conducted at a flow rate of 100 mL/min for 30 min. After sampling, both ends of each sorbent tube were closed with PTFE caps and sealed with parafilm to prevent loosening and external contamination. Following the protocol of Kang et al. [24], the sealed tubes were then placed in glass containers and stored at −80 °C until analysis. All samples were analyzed within 14 days of collection. On the day of analysis, tubes were equilibrated to room temperature for approximately 30 min during autosampler loading, and the maximum time at ambient temperature before thermal desorption did not exceed 3 h. Additional methodological details are provided in Supplementary Text S1.

Chemical analysis

VOC analysis was performed using a thermal desorption-gas chromatography-mass spectrometry (TD-GC-MS) system. Sorbent tubes were thermally desorbed with a TDS-24RD thermal desorption unit (Tatton, China). Primary desorption was conducted at 320 °C for 600 s with a nitrogen flow of 60 mL/min, and analytes were trapped at −10 °C. Secondary desorption was carried out at 300 °C for 60 s, with an injection time of 500 s.

The desorbed compounds were analyzed on a GC2030-TQ8050NX gas chromatograph-mass spectrometer (Shimadzu, Japan) equipped with a DB-WAX capillary column (30 m × 0.25 mm × 0.1 µm; Agilent, USA). The oven temperature program was set as follows: initial temperature of 50 °C (5 min hold), ramped at 6 °C/min to 245 °C, and held for 7.5 min. Helium (99.999% purity) was used as the carrier gas at a constant flow of 1.0 mL/min. The ion source temperature was maintained at 230 °C, and mass spectra were acquired over an *m/z* range of 20–400. A representative TIC chromatogram is provided in Fig. S2 to illustrate peak shapes, retention range, and baseline quality.

Data processing and statistical analysis

Raw chromatographic data were processed using GCMS Solution software (Shimadzu, Japan), including retention time correction, peak alignment, and peak area normalization. Compound identification was performed by comparison with the NIST17 mass spectral library, applying a similarity threshold of > 700.

Subsequent statistical analyses were carried out using Metabo Analyst 6.0 (www.metaboanalyst.ca) and SIMCA 14.1 (Umetrics, Sweden). Hierarchical cluster analysis (HCA), based on Ward's linkage and Euclidean distance, was employed to visualize compound expression patterns as heatmaps. Principal component analysis (PCA) was used to assess global distribution trends and variance structure across samples. Orthogonal partial least squares-discriminant analysis (OPLS-DA) was further applied to identify VOCs that significantly discriminated between experimental groups. Model performance was evaluated using R^2 and Q^2 statistics, and robustness was validated through 200 permutation tests to minimize overfitting risk. Compounds with a variable importance in projection (VIP) score > 1.0 in the partial least squares-discriminant analysis (PLS-DA) models were considered potential discriminatory biomarkers. To account for the systemic influence of corruption timing, a fixed-effects linear model based on the limma package was employed. Overall, PCA and HCA were used to explore global temporal and group patterns, PERMANOVA to test group-level effects, OPLS-DA to build predictive models and rank discriminatory VOCs, and limma-based linear

models to confirm compound-level differences while adjusting for decomposition time.

Results

VOC diversity and abundance

A total of 33 VOCs were identified by GC-MS, encompassing nine chemical classes with the number of compounds indicated in parentheses: alkanes (2), aromatics (1), alcohols (10), ketones (3), aldehydes (1), phenols (3), esters (10), sulfur-containing compounds (2), and nitrogen-containing compounds (1). Minor differences in VOC richness were observed among groups: 30 compounds were detected in the 3-week (3WM) group, 31 compounds in the 5-week (5WM) group, while both the 8-week (8WM) and the 8-week insect-colonized (8WI) groups yielded 33 compounds (Table 1). Among all classes, alcohols and esters were the most abundant in terms of number, each comprising ten compounds.

From the temporal profile of the total peak area, the 3WM and 5WM groups showed a comparable pattern: values increased monotonically over the course of decomposition and reached their maxima on day 20. By contrast, the two groups at 8 weeks of age displayed a rise-then-fall profile, with the non-insect group (8WM) peaking on day 14 and the insect-colonized group (8WI) peaking earlier on day 10, followed by a progressive decline, indicating that insect colonization advanced the timing of the peak (Fig. 1).

With respect to peak area composition (used here as a semi-quantitative measure of relative signal response rather than absolute concentration, given the inherent differences in ionization efficiencies among compounds), sulfur-containing compounds contributed the largest share of the total ion signal in all groups, with dimethyl disulfide (DMDS) consistently providing the highest relative peak area (Fig. 1). In insect-free groups, DMDS increased steadily with time, reaching its maximum on day 20 in the 3WM and 5WM groups but shifting earlier to day 14 in 8WM. In the same insect-free setting, 3-methyl-1-butanol and dimethyl trisulfide (DMTS) ranked second and third, respectively. In these groups, their times of increase differed slightly: 3-methyl-1-butanol rose sharply during the mid-period (days 10–14), whereas DMTS accumulated mainly between days 14 and 20. Under insect colonization, the dynamics of these three compounds changed. In 8WI, DMDS reached its overall maximum by day 10 and exceeded the contemporaneous level in 8WM, demonstrating an earlier and stronger release. Concurrently, the relative share of DMTS increased, surging to its peak on day 10 and moving from third to second rank, while 3-methyl-1-butanol, although elevated, shifted to third. Collectively, these patterns indicate that insect colonization advances and amplifies the expression of sulfur

Table 1 Volatile organic compounds (VOCs) detected during the decomposition process of rats in each experimental group

Compound name	3WM	5WM	8WM	8WI
Tetradecane [21, 25, 26]	✓	✓	✓	✓
Heneicosane	✓	✓	✓	✓
m-Xylene	✓	✓	✓	✓
Dimethyl disulfide [21, 25–30]	✓	✓	✓	✓
Dimethyl trisulfide [21, 25–30]	✓	✓	✓	✓
2-Hexanone [25]	✓	✓	✓	✓
Acetoin	✓	✓	✓	✓
2,5-Hexanedione	✓	✓	✓	✓
Ethyl pentan-2-yl carbonate	✓	✓	✓	✓
Propanoic acid, butyl ester	✓	✓	✓	✓
2-Propenoic acid, butyl ester	✓	✓	✓	✓
Propanoic acid, 2-hydroxy-, ethyl ester [29]	✓	✓	✓	✓
Methyl benzoate [30]	✓	✓	✓	✓
Butanoic acid, 3-methylphenyl ester	✓	✓	✓	✓
Methyl palmitate	✓	✓	✓	✓
Methyl stearate	✓	✓	✓	✓
Dibutyl phthalate	✓	✓	✓	✓
Phthalic acid, hex-3-yl isobutyl ester	✓	✓	✓	✓
1-Butanol [21, 26]	✓	✓	✓	✓
3-Hexanol	✓	✓	✓	✓
2-Hexanol	✓	✓	✓	✓
1-Butanol, 3-methyl- [21, 26, 30]	✓	✓	✓	✓
1-Pentanol [26, 30]	✓	✓	✓	✓
1-Pentanol, 4-methyl- [26]	✓	✓	✓	✓
1-Hexanol [30]	✓	✓	✓	✓
1-Hexanol, 2-ethyl- [26]	✓	✓	✓	✓
(S)-(+)-1,2-Propanediol	✓	✓	✓	✓
2-Phenylethanol [26, 30]	ND	✓	✓	✓
p-Cresol [26, 30]	✓	✓	✓	✓
Phenol [26, 28]	✓	✓	✓	✓
2,4-Di-tert-butylphenol	✓	✓	✓	✓
Benzaldehyde [21, 25, 26, 28]	✓	✓	✓	✓
Oxime-, methoxy-phenyl-	✓	✓	✓	✓

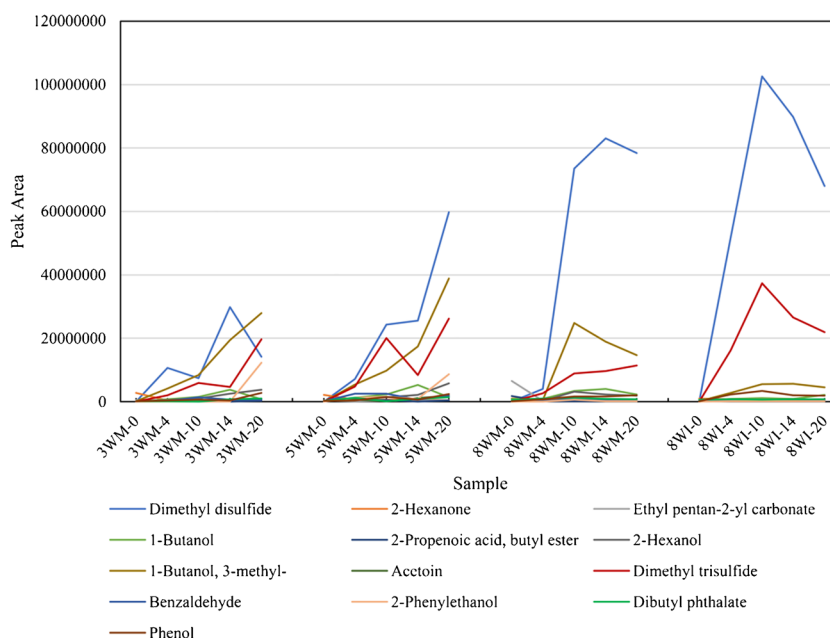
A check mark (✓) indicates that the corresponding VOCs were qualitatively detected (above the identification threshold) in all samples from the three rats within that group at all sampling days (0, 4, 10, 14, and 20 days postmortem). 3WM for 3-week group, 5WM for 5-week group, 8WM for 8-week insect-free group, 8WI for 8-week insect-colonized groups, ND for none detected

compounds (DMDS, DMTS) and reshapes their dominance within the overall VOC profile, rendering sulfur species a more prominent signature of decomposition.

VOC profiles at different decomposition days

Distinct stage-specific combinations of VOCs were observed in all rat groups during decomposition. Under insect-free

Fig. 1 Top 10 volatile organic compounds identified in each experimental group (ranked by relative peak area). For each compound, the value plotted at each sampling day represents the mean relative peak area calculated from the three replicate rats within that group at that time point



conditions, the early stage (day 0) was characterized by 2-hexanone. By day 4, age-dependent differentiation of VOC profiles had emerged: butyl acetate was detected in the 3WM and 5WM groups, whereas dibutyl phthalate appeared in the 5WM and 8WM groups. Concurrently, the levels of aromatic and phenolic compounds, including benzaldehyde and 2,4-di-tert-butylphenol, began to increase. By day 10, all age groups had entered the mid-stage of decomposition, marked by high abundances of aromatic compounds such as benzaldehyde and methoxyphenyl oxime, together with sulfur-containing volatiles including DMDS and DMTS.

Age-related differences became particularly pronounced during the mid-to-late stages (days 14–20). On day 14, the 3WM and 5WM groups were characterized by substantial accumulation of dibutyl phthalate, p-cresol, acetoin, and multiple alcohols, whereas the 8WM group maintained elevated levels of aromatics and sulfur compounds, reflecting a more complex VOC pattern. This divergence was evident in the clustering analysis, where samples from the 8WM group at days 10–14 formed a distinct cluster. By the late stage (day 20), most VOCs in the 3WM and 5WM groups reached peak abundances, while the 8WM group entered a declining phase: aside from phenols, fatty acid methyl esters (methyl palmitate and methyl stearate), and sulfur compounds, the majority of compounds decreased.

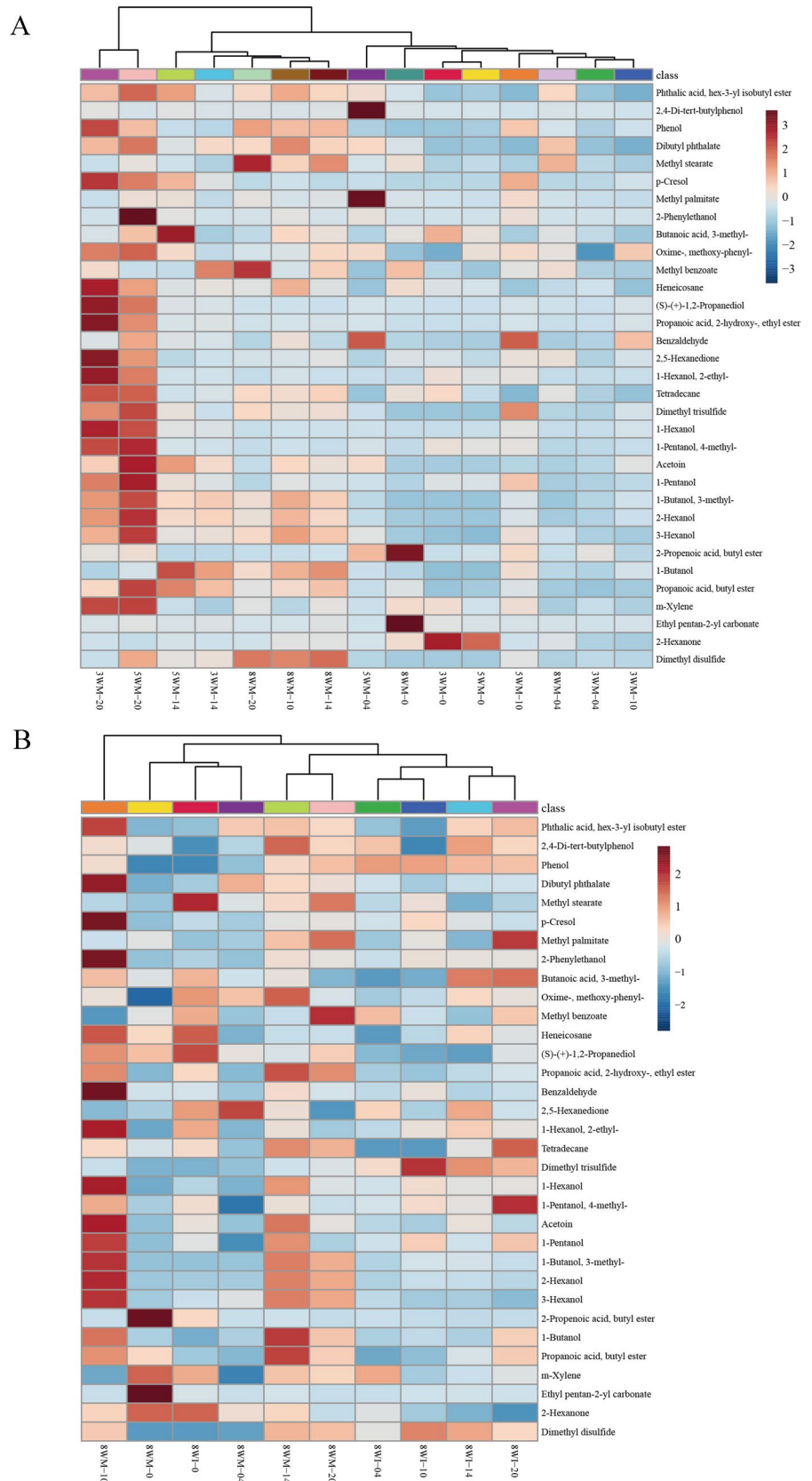
Insect colonization exerted a pronounced accelerating effect on VOC succession. In the insect-colonized group (8WI), large amounts of phenols, DMTS, DMDS, and dibutyl phthalate—typically associated with mid-to-late decomposition—were already detected on day 4, producing a VOC profile closely resembling that of the insect-free 8W group (8WM) at day 10. By day 10, these compounds reached

maximum abundances in the 8WI group, whereas the control group was still in the accumulation phase. Even at day 20, the 8WI group maintained high levels of sulfur compounds and fatty acid derivatives, with consistently greater overall chemical diversity. Hierarchical clustering further confirmed that the temporal succession of VOC profiles in the 8WI group diverged from that of the 8WM group beginning on day 4, with samples from days 10 to 20 forming an independent cluster. This provides robust evidence that insect colonization significantly accelerates the decomposition process.

Hierarchical cluster analysis and heatmaps

HCA of VOC profiles further revealed the influence of age and insect colonization on decomposition chemistry by delineating compound-level expression patterns (Fig. 2). A complementary time-ordered heatmap of the same data is provided in Fig. S3. Among insect-free groups, samples from early (3WM) and mid-aged (5WM) rats clustered closely together during the first 14 days of decomposition, indicating highly similar VOC profile characteristics (Fig. 2A). In sharp contrast, adult rats (8WM) formed a distinct cluster during the mid-stage (days 10–14), diverging significantly from the 3WM/5WM temporal VOC profiles. This separation reflected differences in temporal abundance patterns: whereas the 3WM/5WM groups exhibited substantial accumulation of dibutyl phthalate, p-cresol, and multiple alcohols only in the late stage (days 14–20), the 8WM group already showed pronounced increases in aromatics (e.g., benzaldehyde), sulfur compounds (DMDS, DMTS),

Fig. 2 Hierarchical cluster analysis heatmaps of volatile organic compound (VOC) profiles during rat decomposition. **A** Heatmaps of VOC expression in insect-free groups of different ages (3-week, 3WM; 5-week, 5WM; 8-week insect-free, 8WM). Clustering results show highly similar time-resolved VOC profiles in the 3WM and 5WM groups, whereas the 8WM group diverged markedly during the mid-stage (days 10–14). **B** Heatmaps comparing VOC expression in 8-week insect-free (8WM) and insect-colonized (8-week insect-colonized, 8WI) groups. Clustering highlights the pronounced accelerating effect of insect colonization: the chemical profile of the 8WI group at day 4 closely resembled that of the 8WM group at day 10. Columns are ordered by hierarchical clustering to highlight similarity patterns among samples; a time-ordered version of the heatmap is shown in Fig. S3



and esters during the mid-stage (days 10–14). By day 20, the three age groups converged in the clustering dendrogram.

Insect colonization exerted an even stronger accelerating effect, as highlighted in both the heatmap and clustering results (Fig. 2B). At the onset of decomposition (day 0), the VOC profiles of insect-colonized (8WI) and insect-free (8WM) adult rats were nearly indistinguishable, clustering adjacently. From day 4 onward, however, the 8WI group diverged sharply and, between days 10 and 20, formed an entirely independent cluster. This divergence was driven by the early release (day 4) of typical mid-to-late stage markers—including DMDS, DMTS, p-cresol, and dibutyl phthalate—at levels comparable to those detected in the insect-free group at day 10.

Principal component analysis

PCA was used to evaluate the overall effects of time, age, and insect colonization on VOC variation. In the age-focused model (Fig. 3A), the first two principal components (PC1 and PC2) together explained 51.2% of the total variance, with PC1 accounting for 38.5% and PC2 for 12.7%. PC1 correlated strongly with decomposition time, as all groups (3WM, 5WM, 8WM) exhibited a clear temporal succession from day 0 (negative axis) to day 20 (positive axis). Age-related effects were also evident: biological replicates from the 3WM and 5WM groups clustered tightly

at each time point, indicating high intra-group consistency. In contrast, 8WM samples traced a distinct temporal path that was clearly displaced from the younger groups, with the divergence in time-resolved VOC profiles particularly pronounced by day 20.

In the insect colonization model (Fig. 3B), PC1 and PC2 explained 29.3% and 13.4% of the variance, respectively. Here, the accelerating effect of insects was clearly visualized. At day 0, VOC profiles of insect-colonized (8WI) and insect-free (8WM) groups overlapped, showing no distinction. From day 4 onward, however, their temporal succession patterns rapidly diverged, with 8WI samples migrating along PC1 much faster than 8WM. Its position in the principal component space on day 4 had already approached the level of the same group on day 10. Between days 10 and 20, 8WI formed an entirely separate temporal succession pattern in PCA space, with greater dispersion.

Pairwise PERMANOVA based on 999 permutations further supported these conclusions (Table 2). Across the full period (d0–d20), the age effect was not significant ($F = 1.7768$, $R^2 = 0.078$, $p = 0.143$), likely due to convergence of profiles at the start (day 0) and end (day 20). Restricting the analysis to days 0–14, age differences became significant ($F = 4.2322$, $R^2 = 0.204$, $p = 0.005$), explaining 20.4% of the variance. Pairwise comparisons revealed significant separation between 3WM vs 8WM ($F = 6.687$, $R^2 = 0.2331$, $p = 0.021$) and 5WM vs 8WM ($F = 5.2975$,

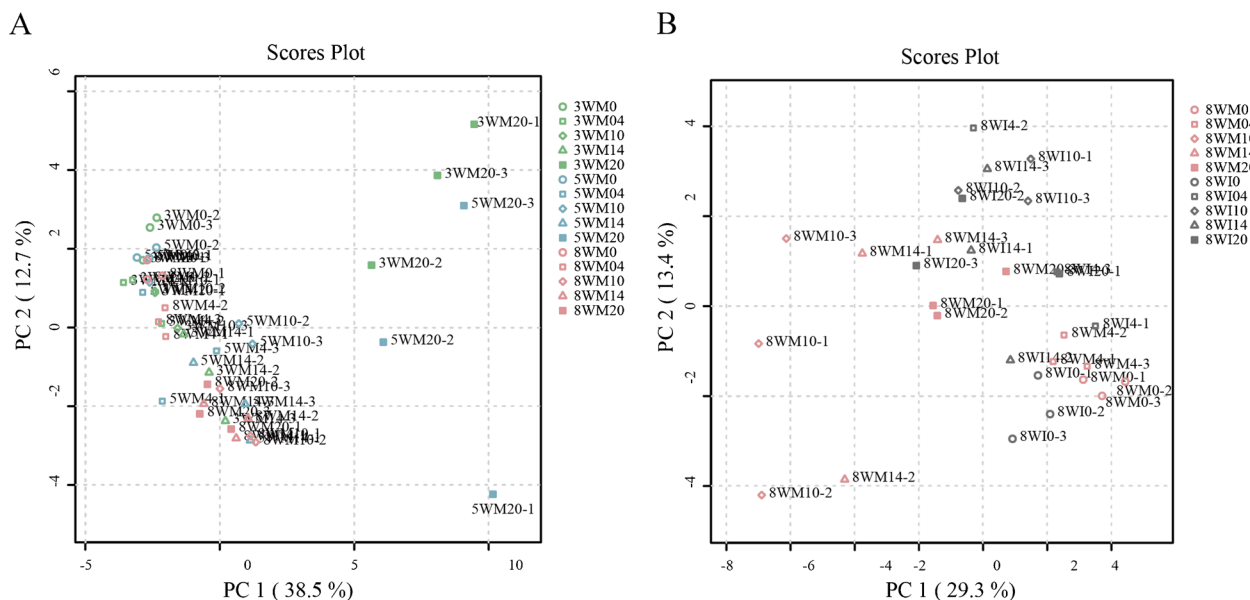


Fig. 3 Principal component analysis score plots of volatile organic compound (VOC) profiles during rat decomposition. **A** Comparison of insect-free groups of different ages (3-week, 3WM; 5-week, 5WM; 8-week insect-free, 8WM). The PCA score plot shows that decomposition time is the main source of variation, with samples from all groups following a clear temporal progression along principal component 1 (PC1). The 3WM and 5WM groups exhibit highly similar

time-resolved VOC profiles, whereas the 8WM group follows a distinct temporal pattern that is clearly separated from the 3WM and 5WM groups, with replicates clustering tightly at each time point. **B** Comparison of 8-week insect-free (8WM) and insect-colonized (8-week insect-colonized, 8WI) groups. The model illustrates the strong accelerating effect of insect colonization, with 8WI samples at day 4 already positioned close to the same group samples at day 10

Table 2 Pairwise permutational multivariate analysis of variance (PERMANOVA) results

Groups	Time window	F. Model	R^2	p val (based on 999 permutations)
3WM/5WM/8WM	d0–d20	1.7768	0.07801	0.143
3WM/5WM/8WM	d0–d14	4.2322	0.20414	0.005
3WM/8WM	d0–d14	6.687	0.2331	0.021
5WM/8WM	d0–d14	5.2975	0.19407	0.021
3WM/5WM	d0–d14	1.547	0.0657	0.226
8WM/8WI	d4–d14	5.7932	0.26583	0.012
8WM/8WI	d4–d20	6.3587	0.22422	0.002

3WM for 3-week group, 5WM for 5-week group, 8WM for 8-week insect-free group, 8WI for 8-week insect-colonized groups

$R^2=0.1941$, $p=0.021$), but not between 3WM vs 5WM ($F=1.547$, $R^2=0.0657$, $p=0.226$), indicating that the age effect was mainly driven by divergence of the 8WM group from the younger cohorts.

Similarly, for insect colonization, no significant group-level differences were observed across the full 0–20-day period ($F=3.2555$, $R^2=0.104$, $p=0.060$), again reflecting similarity at day 0 and convergence at later stages. When day 0 was excluded (d4–d20), the effect became significant ($F=6.3587$, $R^2=0.224$, $p=0.002$), with insect colonization explaining 22.4% of the variance. This confirms that from day 4 onward, insect activity exerted a consistent and substantial influence on VOC composition.

Orthogonal partial least squares-discriminant analysis

To identify key chemical markers underlying group differences, OPLS-DA models were constructed (Table 3; Table S1). For age effects, the overall model comparing early/mid-stage groups (3WM/5WM) with adults (8WM) demonstrated strong discriminative power, with parameters of $R^2X=0.686$, $R^2Y=0.677$, and predictive ability $Q^2=0.414$ (Table 3). Cross-validation by CV-ANOVA confirmed model significance ($p<0.05$). The score plot (Fig. 4A) clearly showed separation of the 8WM group from the combined 3WM/5WM group, while 3WM and 5WM samples overlapped extensively.

Pairwise models provided further resolution. The 3WM vs 8WM comparison yielded excellent performance ($R^2X=0.766$, $R^2Y=0.936$, $Q^2=0.749$, $p<0.05$) (Table 3), achieving complete separation of the two groups (Fig. 4B). Similarly, the 5WM vs 8WM model performed well ($R^2X=0.612$, $R^2Y=0.895$, $Q^2=0.612$, $p<0.05$), with clear group separation (Fig. 4C). In contrast, the 3WM vs 5WM model showed weak discrimination and poor predictive power ($Q^2=0.244$, $p=0.13$), consistent with previous analyses. For all significant models, 200 permutation tests yielded negative Q^2 intercepts, confirming model robustness and excluding overfitting (Fig. 4E–G).

For insect colonization effects, the OPLS-DA model also showed strong discriminative ability, effectively separating insect-colonized (8WI) from insect-free (8WM) carcasses. The model parameters were $R^2X=0.469$, $R^2Y=0.796$, and $Q^2=0.555$, with significance confirmed by CV-ANOVA ($p<0.05$) (Table 3). These findings were consistent with the PERMANOVA results ($R^2\approx0.224$, $p=0.002$). The score plot (Fig. 4D) displayed two spatially distinct clusters. Corresponding permutation tests (Fig. 4H) further validated the stability and reliability of the model, ruling out the possibility of random fitting.

Based on the validated OPLS-DA models, compounds with the highest discriminative contributions were identified using VIP scores, with a threshold set at >1.0 (Table S2; Table S3). In the overall model distinguishing adults (8WM) from younger groups (3WM/5WM),

Table 3 Orthogonal partial least squares (OPLS) model results

Group	R^2X	R^2Y	Q^2	CV-ANOVA F	CV-ANOVA p	Permutation Q^2_{int}
3WM/5WM/8WM	0.686	0.677	0.414	3.2067	0.000176911	−0.536
3WM/5WM	0.546	0.424	0.244	1.93261	0.136208	−0.277
3WM/8WM	0.766	0.936	0.749	5.78724	0.000530369	−0.99
5WM/8WM	0.612	0.895	0.612	4.02324	0.00493127	−1.04
8WM/8WI	0.469	0.796	0.555	4.75126	0.00276332	−0.834

3WM for 3-week group, 5WM for 5-week group, 8WM for 8-week insect-free group, 8WI for 8-week insect-colonized groups

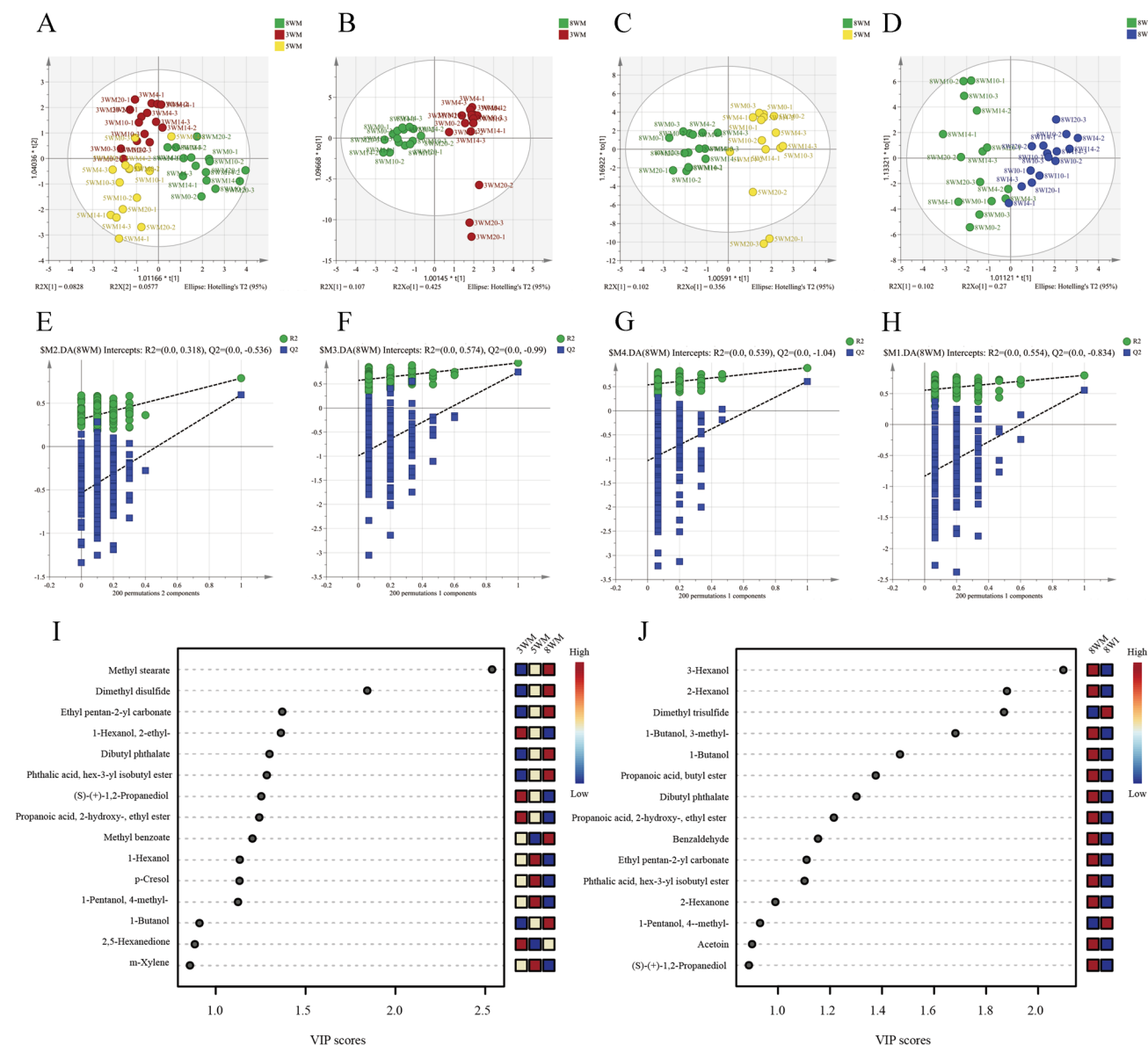


Fig. 4 Orthogonal partial least squares-discriminant analysis (OPLS-DA) score plots, permutation tests, and variable importance in projection (VIP)-selected compounds of volatile organic compound (VOC) profiles during rat decomposition. **A, E** Overall OPLS-DA model comparing early/mid-stage (3-week, 3WM; 5-week, 5WM) and adult (8-week insect-free, 8WM) groups: score plot and corresponding permutation test. **B, F** Pairwise OPLS-DA model for 3-week (3WM) vs. 8-week insect-free (8WM) groups: score plot and permutation

test. **C, G** Pairwise OPLS-DA model for 5-week (5WM) vs. 8-week insect-free (8WM) groups: score plot and permutation test. **D, H** OPLS-DA model comparing 8-week insect-free (8WM) and insect-colonized (8-week insect-colonized, 8WI) groups: score plot and permutation test. **I** VOCs with high contributions in the overall age model (3WM/5WM/8WM). **J** VOCs with high contributions in the insect colonization model (8WM/8WI)

the top markers included methyl stearate, DMDS, and 1-hexanol, 2-ethyl- (Fig. 4I). In the insect colonization model, the most influential compounds were dominated by alcohols—such as 3-hexanol and 2-hexanol—together with DMTS (Fig. 4J).

Linear models with covariate adjustments

To further validate the independent effects of age and insect colonization after controlling for decomposition time, we applied linear models with sampling day as a covariate and

Table 4 Differential compounds derived from linear models with covariate adjustments

Group	Compound name	CAS	Formula	Retention time (min)	VIP score	logFC	adj. <i>p</i>
3WM/5WM/8WM	Methyl stearate	112-61-8	C19H38O2	32.57	2.5381	1.4997	1.18E-05
	Dimethyl disulfide	624-92-0	C2H6S2	5.66	1.8434	1.0939	0.001642
	Ethyl pentan-2-yl carbonate	NA	C8H16O3	7.11	1.3703	0.76562	0.04886
	Methyl benzoate	93-58-3	C8H8O2	19.12	1.2044	0.72184	0.04886
8WM/8WI	3-Hexanol	623-37-0	C6H14O	9.28	2.0969	-1.1446	0.010392
	2-Hexanol	626-93-7	C6H14O	9.68	1.8801	-1.0262	0.010611
	Dimethyl trisulfide	3658-80-8	C2H6S3	13.74	1.8689	1.0201	0.010611
	1-Butanol, 3-methyl-	123-51-3	C5H12O	9.44	1.6821	-0.91816	0.021718

3WM for 3-week group, 5WM for 5-week group, 8WM for 8-week insect-free group, 8WI for 8-week insect-colonized groups, NA for not available

corrected for multiple comparisons using the false discovery rate (FDR) (Table 4; Table S4; Fig. 5A, B).

The results confirmed significant age-associated differences, primarily reflecting systematic divergence between the adult (8WM) group and the younger cohorts (3WM/5WM). After adjusting for time, numerous discriminatory compounds were identified between 8 and 3WM/5WM. The 8WM group was characterized by significant enrichment

of methyl stearate and DMDS, along with elevated levels of methyl benzoate, ethyl pentan-2-yl carbonate, and dibutyl phthalate. By contrast, the 3WM/5WM groups showed relatively higher abundances of several primary alcohols (e.g., 1-pentanol, 2-phenylethanol) and certain phenolic and aromatic compounds (e.g., *p*-cresol, benzaldehyde) (all FDR-*p* < 0.05). Figure 5A highlights the shared discriminatory compounds between 8 and 3WM/5WM. Importantly,

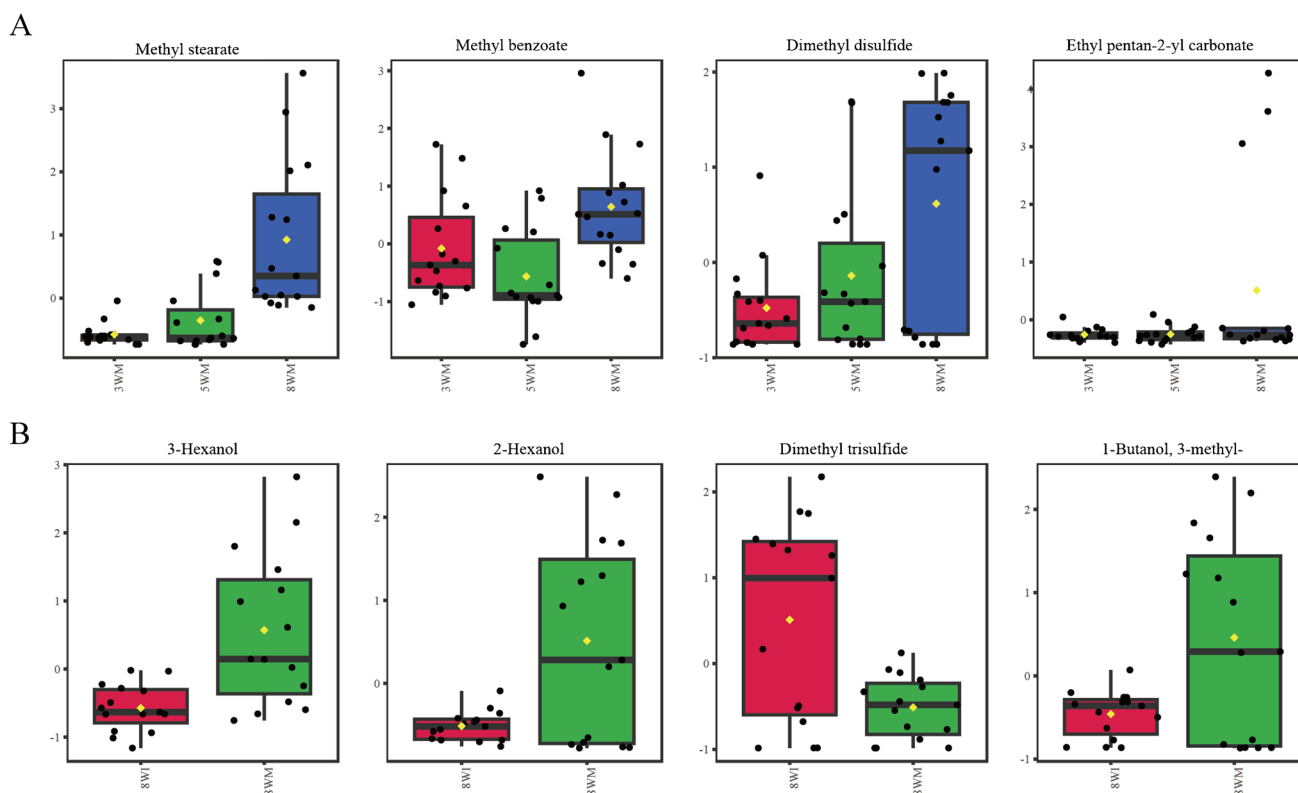


Fig. 5 Boxplots comparing the relative abundances of key discriminatory volatile organic compounds. **A** Representative compounds differentiating adults (8-week insect-free, 8WM) from younger groups

(3-week, 3WM; 5-week, 5WM). **B** Representative compounds differentiating insect-colonized (8-week insect-colonized, 8WI) from insect-free (8WM) carcasses

no compounds passed FDR correction between 3 and 5WM (FDR- $p \geq 0.05$).

Similarly, the independent effect of insect colonization was validated after time adjustment. Even after controlling for sampling day, significant differences persisted between insect-colonized (8WI) and insect-free (8WM) carcasses. Notably, DMTS was consistently enriched in 8WI, whereas alcohols such as 3-hexanol, 2-hexanol, and 3-methyl-1-butanol were significantly higher in 8WM.

Discussion

This study systematically delineated the dynamic succession of volatile organic compounds (VOCs) during rat decomposition within a multilevel statistical framework. PCA established time as the dominant axis of variation in VOC profiles, while HCA showed that age and insect colonization modulate decomposition via distinct mechanisms—age by increasing chemical complexity of VOC profiles and insects by accelerating the temporal succession of decomposition VOC profiles. Supervised OPLS-DA further confirmed pronounced differences between adults (8W) and younger cohorts (3W/5W), demonstrated a strong accelerating effect of insect colonization, and identified the key chemical markers driving these separations. Collectively, these findings deepen the understanding of decomposition-associated VOC profiles and provide molecular evidence to improve PMI estimation.

The association between VOC profiles and time since death was confirmed, and the stage-specific signatures aligned closely with established thanatochemistry and previous VOC studies on decomposing human and animal remains [4, 11, 26]. The early stage was characterized by 2-hexanone, plausibly arising from lipid oxidation [31]. The mid-stage was defined by pronounced accumulation of aromatics (e.g., benzaldehyde) and sulfur volatiles (DMDS; DMTS), which in our data reached high abundances by day 10, indicating vigorous anaerobic protein degradation—particularly of sulfur-containing amino acids [13, 32]. In the late stage, alcohols and esters showed distinct temporal trajectories among age groups: in 3WM/5WM, their relative abundances continued to rise and many compounds reached their maxima by day 20, whereas in 8WM, most alcohols and simple esters had already started to decline, with only fatty acid methyl esters (methyl palmitate and methyl stearate) remaining prominent, a pattern consistent with fatty acid esterification and age-related shifts in microbial metabolism [26, 33]. The occurrence and mid-to-late stage prominence of these compounds is also in line with recent rat decomposition VOC studies. Martin et al. and Malevic et al. characterized, respectively, the headspace above submerged rat cadavers and the dissolved volatilome by DI-SPME, and

in both studies, sulfur compounds (notably DMDS and DMTS), benzaldehyde, 2-hexanone, and several carboxylic acids emerged as recurrent components of the rat cadaver volatilome [25, 27]. Despite clear differences in matrix and environment, the repeated detection and mid-to-late stage dominance of these same compounds in our terrestrial rat model support a consistent rat-specific volatilome signature under our experimental conditions. Multivariate analyses (PCA, HCA, OPLS-DA) further demonstrated a continuous and ordered temporal succession of VOC profiles over time.

Age emerged as a key determinant of VOC variability. Adult rats (8WM) exhibited greater dispersion and higher chemical diversity during the mid-stage compared to younger groups (3WM/5WM). The HCA heatmap showed that the 8W group formed a distinct cluster at days 10–14, diverging markedly from the time-resolved VOC profiles of the 3W/5W groups—indicative of a faster decomposition course and a more complex biochemical decomposition pathways. Consistently, PCA score plots revealed greater scatter for 8W samples in principal component space, underscoring age as a major driver of complexity of VOC compositions. Furthermore, time-adjusted linear models detected no significant differential compounds between 3 and 5W, confirming their highly similar VOC patterns during decomposition, while both differed significantly from 8W. This divergence is mechanistically can be mechanistically explained by age-related differences in tissue composition and postmortem microbiota. Across developmental stages, mammals differ in tissue composition (e.g., skin architecture, lipid content) and in their initial thanatomicrobiome [34, 35]. Recent large-scale human decomposition studies indicate that a relatively conserved interdomain microbial network underpins cadaver breakdown, while host and environmental factors mainly modulate taxon abundances and activity [36]. Accordingly, we interpret the greater dispersion and richness of VOC profiles in adult rats as a consequence of age-related differences in biochemical substrates acting on this shared decomposer community, rather than the recruitment of distinct microbial assemblages. These findings are consistent with recent reports linking age to VOC emission patterns in humans [20, 37]. Forensic models based on VOCs must therefore incorporate age as a mechanistically grounded, predictable, and correctable parameter.

Finally, the impact of insect colonization on the evolution of VOC profiles is direct and highly pronounced. This study demonstrates that insect activity greatly accelerates the decomposition process. The most intuitive evidence is that, in our experiment, the VOC profile of the insect group (8WI) on day 4 closely resembled that of the insect-free control (8WM) on day 10, indicating that the temporal succession of VOC profiles advanced by approximately 6 days. PCA likewise showed that 8WI scores migrated along PC1 much more rapidly than those of 8WM, providing a direct

visualization of the advance in decomposition. More importantly, insect activity not only “accelerated” decomposition but also “remodeled” the chemical composition of VOCs: in the 10–20-day window, HCA revealed a branch entirely independent of 8WM, confirming long-lasting and marked changes in biochemical decomposition pathways induced by insect activity and in line with previous observations that necrophagous insects can qualitatively alter cadaveric volatiles [13, 21, 38–40]. Taken together, these findings highlight the multifaceted influence of insects on cadaver chemistry, expressed through several interacting processes: large maggot masses can locally raise body temperature, thereby further accelerating all biochemical reactions [41, 42]; insects also carry and disseminate exogenous microorganisms, altering the existing postmortem microbiome [43, 44]. Heavy larval feeding directly removes substrates available for microbial decomposition and, via excretions/secretions, releases peptides and enzymes with broad-spectrum antimicrobial activity, thereby reducing local bacterial load and altering microbial catabolic pathways and gas/odor production capacity [45, 46]. When microbial substrates are rapidly consumed and the bacterial community is suppressed, the relative rates and timing of volatiles driven by proteolysis and sulfur-containing amino acid catabolism (e.g., DMDS/DMTS), as well as intermediate alcohols (e.g., 3-methyl-1-butanol), may shift toward earlier occurrence or show amplification/suppression-consistent with the earlier DMDS/DMTS peaks observed in 8WI. Necrophagous flies also exhibit feeding preferences among tissues (e.g., viscera, muscle, subcutaneous fat), and different tissues possess distinct volatiles; thus, preferential larval consumption of particular tissues can systematically reshape VOC temporal patterns by altering the relative availability of biochemical precursors [47–49]. Therefore, cadavers with insect colonization should be regarded as biochemically distinct systems, and this qualitative change must be fully accounted for when constructing forensic models.

From a methodological standpoint, this study integrates multiple complementary statistical strategies to ensure robust inference aligning with and extending previous workflows demonstrated by Stefanuto et al. in human decomposition studies [16]. Unsupervised PCA and HCA were first used to visualize the intrinsic structure of the data without imposing class labels, an approach widely recommended in metabolomics for delineating global trends and natural clustering patterns [50]. To specifically extract variation associated with our experimental factors (age and insect colonization) and to filter out unrelated orthogonal variation, we then applied OPLS-DA, following the framework originally proposed by Trygg and Wold [51], which improves interpretability by explicitly separating predictive from orthogonal components. The resulting models showed acceptable predictive ability (e.g., $Q^2 > 0.5$), indicating that the detected

group differences were not driven by overfitting. Finally, we used limma-based linear models with empirical Bayes variance moderation, which are well suited to high-dimensional, small-sample designs, to perform hypothesis testing while controlling for decomposition time as a covariate [52]. This confirmed that age and insect colonization act as independent drivers of VOC variance rather than mere byproducts of differing decomposition rates. Together, this multilayered analytical framework provides a coherent and reliable paradigm for forensic volatilomics research.

From an instrumentation standpoint, our sampling/analysis stack (sorbent tubes + thermal desorption GC-MS) aligns with mainstream, field-deployable volatilomics platforms. TD-GC-MS is widely used across breath, microbial, and decomposition studies, with demonstrated analytical performance and successful application to human-remains VOCs—including aquatic contexts—supporting transferability to real forensic scenarios [53–55]. Although species-specific differences in anatomy, microbiota, and environment will affect absolute VOC outputs, the dominant chemical classes observed here—particularly sulfur compounds and esters—are in line with established human decomposition VOC databases [12]. These parallels suggest that our workflow could be adapted in future studies on human cadavers or large animal models; however, the present rat data do not demonstrate such transferability. In particular, whether explicit modelling of age and insect activity as covariates improves VOC-based PMI models in human material will need to be tested empirically in dedicated human or large-animal studies. Successful implementation will, however, require appropriate chemometric expertise and, more importantly, larger, externally validated datasets covering diverse environments, body conditions, and insect communities. Within this context, the age- and insect-associated VOCs identified in the present study should be regarded as preliminary candidates that may inform, but do not yet define, simplified marker panels for future routine applications.

Despite the strength of these conclusions, several limitations merit consideration. First, the experiments were performed under tightly controlled laboratory conditions in a rat model, which may limit extrapolation to complex, heterogeneous natural environments—particularly when extending findings to human cadavers. Second, the sample size was relatively limited, and only a single insect species (*S. pergrina*) was examined, leaving the effects of multi-species interactions unaddressed. Third, the study focused exclusively on VOC profiling, without integration of microbial, metabolomic, or other molecular datasets, which restricts a more comprehensive understanding of the mechanistic underpinnings of decomposition.

From an analytical standpoint, additional constraints also apply. VOCs were analyzed from Tenax TA tubes stored at $-80\text{ }^{\circ}\text{C}$ rather than immediately after sampling. Although

this practice is widely used and supported by previous stability studies on Tenax-based sorbent tubes, some loss of highly labile analytes cannot be entirely excluded; however, any such effects are expected to act systematically across all treatment groups. The PTFE sampling bags, polypropylene containers, and sterile sand used as support matrices were not quantitatively characterized for blank emissions or sorptive behavior and may act as weakly emissive and/or adsorptive media, reducing the absolute headspace abundance of some compounds. Because the same container–sand system was applied uniformly and randomly to all age and insect treatments, these matrix effects are expected to be largely systematic rather than group-specific, and our results should be interpreted as headspace-accessible VOC profiles within a standardized container/sand microenvironment rather than as absolute emission fluxes. Furthermore, in the insect-colonized group, the VOC profiles represent the integrated output of the decomposing carcass, its microbiome, and *S. peregrina* larvae, including insect-associated microorganisms. Without “fly-only” controls, compounds enriched in 8WI should therefore be regarded as markers of the carcass–insect system rather than exclusively cadaver-derived VOCs. Finally, our VOC panel predominantly covers C_4 – C_{18} compounds, and very light, highly volatile species are likely under-represented, a systematic bias that should be considered when comparing our data with studies using non-polar columns and sorbents optimized for C_1 – C_3 compounds.

From a forensic perspective, these findings translate into three main conclusions. First, time-structured VOC profiles can function as a chemical timing signal for PMI estimation, as they follow a continuous and ordered succession with stage-specific markers from ketones to aromatics/sulfur compounds and, finally, alcohols and esters. Second, age acts as a key intrinsic modifier of this process: different developmental stages do not merely differ in absolute abundance but in the shape and timing of their VOC time courses, with adults showing greater mid-stage chemical diversity and within-group dispersion than younger animals. Third, insect colonization represents a distinct extrinsic scenario in which the VOC time window is reshaped and effectively shortened, with mid-stage markers appearing earlier and decaying sooner than in insect-free carcasses. Taken together, these results indicate that any VOC-based PMI inference should, in practice, account for decomposition time, age strata, and the presence or absence of insect activity, rather than assuming a single, universal calibration curve for all cadavers.

To overcome these limitations and advance practical applications, future research should proceed along three directions: (i) development of multimodal predictive models by integrating VOCs with microbiome and metabolomic data to enhance the precision and robustness of PMI estimation; (ii) extension of experimental systems to human cadavers or large animal models that better approximate real

forensic scenarios, thereby strengthening translational value; and (iii) innovation in portable VOC collection and rapid detection technologies to enable the transition from laboratory-based analyses to field-ready forensic tools. Through deep interdisciplinary collaboration and the integration of cutting-edge analytical technologies, it will be possible to establish a more accurate, reliable, and operationally relevant PMI estimation framework, thereby advancing the scientific rigor and standardization of forensic practice.

Conclusions

This study, conducted in a rat model, systematically delineates the time-series characteristics of volatile organic compounds (VOCs) during decomposition and the effects of developmental stage and insect colonization on their dynamic evolution. First, the temporal succession of decomposition VOC profiles is tightly associated with PMI and exhibits clear stage-specific chemical signatures: ketones dominate early, aromatics and sulfur volatiles accumulate in the mid phase, and a shift toward alcohols and esters characterizes the late phase. Against this backdrop, age functions as a key intrinsic factor that reshapes the form of the time-dependent VOC profiles: developmental stages differ not only in abundance but also in profile geometry and peak timing, with adult carcasses (8WM) showing greater chemical diversity and within-group dispersion during the mid stage. By contrast, insect colonization, as a critical extrinsic factor, markedly remodels, advances, and truncates the temporal succession of VOC profiles: mid-stage markers appear earlier and decay sooner, compressing the effective temporal window and bringing forward the termination of late-stage features. Together, these results provide new molecular evidence for PMI estimation and emphasize that decomposition time, age-related physiology, and insect colonization must all be explicitly considered when using VOCs as temporal markers. In practical terms, future VOC-based PMI models should incorporate age as a covariate and implement separate “with-insect” and “insect-free” branches, so as to reduce systematic bias and improve the robustness and forensic utility of volatilomics-derived estimates.

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Author contribution Chengxin Ye: conceptualization, methodology, writing—original draft preparation, writing—review and editing, visualization. Yang Xia: validation, writing—original draft preparation, writing—review and editing. Xiangyan Zhang: conceptualization, methodology, writing—review and editing. Jian Zhao: writing—review

and editing, formal analysis. Sheng Hu: resources, writing—review and editing. Fan Yang: resources, writing—review and editing. Sile Chen: validation, investigation. Hai Wu: validation, investigation. Lei Zhao: writing—review and editing, visualization. Xingchun Zhao: writing—review and editing, supervision, funding acquisition. Yadong Guo: conceptualization, writing—review and editing, supervision, funding acquisition.

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Data availability The data presented in the manuscript and Supplementary Material is available upon request.

Declarations

Ethics approval Approval was obtained by the Institutional Review Board (IRB) of the School of Basic Medical Sciences, Central South University (2023-KT085).

Consent to participate Not applicable.

Consent for publication Not applicable.

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